

Project Design Phase-II Technology Stack (Architecture & Stack)

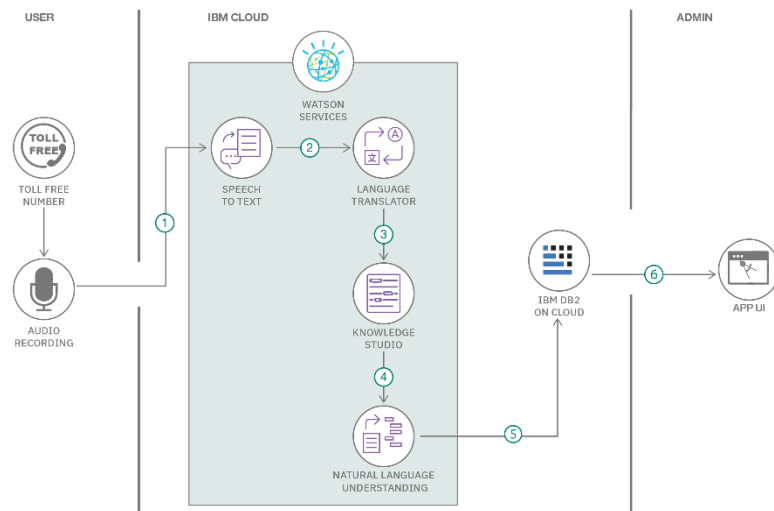
Date	31 January 3035
Team ID	
Project Name	
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Order processing during pandemics for offline mode

Reference: <https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/>



Guidelines:

- Include all the processes (As an application logic / Technology Block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third party API's etc.)
- Indicate Data Storage components / services
- Indicate interface to machine learning models (if applicable)

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Responsive web application for users and administrators	HTML5, CSS3, JavaScript, React.js, Bootstrap
2.	Application Logic-1	User Authentication, Registration and Login	Node.js, Express.js
3.	Application Logic-2	Meal Planning and Nutrition Tracking	Node.js, Express.js
4.	Application Logic-3	Dashboard Analytics and Report Generation	React.js, Chart.js
5.	Database	Stores users, meal plans, nutrition records and progress data	MongoDB, Mongoose
6.	Cloud Database	Cloud-hosted database for secure storage	MongoDB Atlas
7.	File Storage	Stores profile images and uploaded files	Local File System / Cloudinary (Optional)
8.	External API-1	Nutrition and food information API	USDA FoodData Central API / Edamam Nutrition API
9.	External API-2	Email verification and notifications	EmailJS / Nodemailer
10.	Machine Learning Model	AI-powered nutrition recommendations (Optional Enhancement)	TensorFlow.js / OpenAI API
11.	Infrastructure (Server / Cloud)	Application deployment and hosting	Local Server, Render, Vercel, Netlify

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Frontend and backend frameworks used in the application	React.js, Node.js, Express.js, Bootstrap, Mongoose
2.	Security Implementations	Secure login, password encryption, authentication and authorization	JWT Authentication, bcrypt Password Encryption, HTTPS, Input Validation, CORS
3.	Scalable Architecture	Three-tier architecture supporting modular and scalable development	MERN Stack, RESTful APIs, MongoDB
4.	Availability	Ensures continuous availability through cloud hosting and database services	MongoDB Atlas, Render, Vercel
5.	Performance	Optimized application performance using efficient database queries, API optimization and frontend optimization	MongoDB Indexing, Axios, React Lazy Loading, Code Splitting, Express Middleware

References:

<https://c4model.com/>

<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>